CentraleSupélec - MVA
Large Scale and Distributed Optimization
Part V.2: Douglas-Rachford algorithm

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Optimization strategies

Motivation

Let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$. We want to

$$\underset{x \in \mathcal{H}}{\text{minimize}} \;\; f(x) + g(x).$$

Possible solutions :

- gradient descent algorithm $\Rightarrow$ $f + g$ needs to be smooth
- proximal point algorithm $\Rightarrow$ $f + g$ needs to be "proximable"
- Forward-Backward algorithm $\Rightarrow$ g needs to be smooth

Can we find a splitting algorithm when both f and g are nonsmooth?

Reflection of the prox

Let $\gamma \in]0, +\infty[$ and let $f \in \Gamma_0(\mathcal{H})$.
The reflection of the proximity operator defined as

$$\mathrm{rprox}_{\gamma f} = 2\mathrm{prox}_{\gamma f} - \mathrm{Id}$$

is nonexpansive.

Fixed points

Let $\gamma \in]0, +\infty[$, let $f \in \Gamma_0(\mathcal{H})$ and let $g \in \Gamma_0(\mathcal{H})$.
We have

$$\operatorname{zer}(\partial f + \partial g) = \operatorname{prox}_{\gamma g}(\operatorname{Fix} T)$$

where $T = \operatorname{rprox}_{\gamma f} \circ \operatorname{rprox}_{\gamma g}$.

Fixed points

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We have

$$\operatorname{zer}(\partial f + \partial g) = \operatorname{prox}_{\gamma g}(\operatorname{Fix} T)$$

where $T = \operatorname{rprox}_{\gamma f} \circ \operatorname{rprox}_{\gamma g}$.

Proof: Let $x \in \mathcal{H}$.

$$\begin{aligned}
0 \in \gamma\big(\partial f(x) + \partial g(x)\big) &\Leftrightarrow (\exists y \in \mathcal{H})\ \ x - y \in \gamma \partial f(x) \text{ and } y - x \in \gamma \partial g(x) \\
&\Leftrightarrow (\exists y \in \mathcal{H})\ \ 2x - y \in (\operatorname{Id} + \gamma \partial f)x \\
&\qquad\qquad \text{and } x = \operatorname{prox}_{\gamma g}(y) \\
&\Leftrightarrow (\exists y \in \mathcal{H})\ \ x = \operatorname{prox}_{\gamma f}\big(\operatorname{rprox}_{\gamma g}(y)\big) \text{ and } x = \operatorname{prox}_{\gamma g}(y) \\
&\Leftrightarrow (\exists y \in \mathcal{H})\ \ \operatorname{rprox}_{\gamma f}\big(\operatorname{rprox}_{\gamma g}(y)\big) = 2x - \operatorname{rprox}_{\gamma g}(y) = y \\
&\qquad\qquad \text{and } x = \operatorname{prox}_{\gamma g}(y) \\
&\Leftrightarrow (\exists y \in \operatorname{Fix} T)\ \ x = \operatorname{prox}_{\gamma g}(y).
\end{aligned}$$

Douglas-Rachford algorithm

Let $\mathcal{H}$ be a Hilbert space.

Let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$.

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ z_n = \operatorname{prox}_{\gamma f}(2y_n - x_n) \\ x_{n+1} = x_n + \lambda_n(z_n - y_n). \end{cases}$$

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Let $\gamma \in]0, +\infty[$ and let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[0, 2]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(2 - \lambda_n) = +\infty$.

We assume that $\operatorname{zer}(\partial f + \partial g) \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ z_n = \operatorname{prox}_{\gamma f}(2y_n - x_n) \\ x_{n+1} = x_n + \lambda_n(z_n - y_n). \end{cases}$$

The following properties are satisfied:

1. $x_n \rightharpoonup \widehat{x}$
2. $\widehat{y} = \operatorname{prox}_{\gamma g}\widehat{x} \in \operatorname{Argmin}(f + g)$
3. $z_n - y_n \to 0$.

Douglas-Rachford algorithm

Proof: Let $T = \mathrm{rprox}_{\gamma f} \circ \mathrm{rprox}_{\gamma g}$. T is nonexpansive and $\varnothing \neq \mathrm{zer}\,(\partial f + \partial g) = \mathrm{prox}_{\gamma g}(\mathrm{Fix}\,T) \Rightarrow \mathrm{Fix}\,T \neq \varnothing$.
Moreover, for every $n \in \mathbb{N}$,

$$
\begin{aligned}
x_{n+1} &= x_n + \lambda_n\big(\mathrm{prox}_{\gamma f}(2\mathrm{prox}_{\gamma g}(x_n) - x_n) - \mathrm{prox}_{\gamma g}(x_n)\big) \\
&= x_n + \frac{\lambda_n}{2}\big(2\mathrm{prox}_{\gamma f}(2\mathrm{prox}_{\gamma g}(x_n) - x_n) - 2\mathrm{prox}_{\gamma g}(x_n) + x_n - x_n\big) \\
&= x_n + \frac{\lambda_n}{2}\big(2\mathrm{prox}_{\gamma f}(\mathrm{rprox}_{\gamma g}(x_n)) - \mathrm{rprox}_{\gamma g}(x_n) - x_n\big) \\
&= x_n + \frac{\lambda_n}{2}\big(Tx_n - x_n\big).
\end{aligned}
$$

$\Rightarrow$ Krasnosel'skii-Mann algorithm with relaxation factors $(\lambda_n/2)_{n\in\mathbb{N}}$.
We deduce that $Tx_n - x_n \to 0$ and $x_n \rightharpoonup \widehat{x} \in \mathrm{Fix}\,T$.

Douglas-Rachford algorithm

Proof: For every $n \in \mathbb{N}$,

$$z_n - y_n = \operatorname{prox}_{\gamma f}(2\operatorname{prox}_{\gamma g}(x_n) - x_n) - \operatorname{prox}_{\gamma g}(x_n) = \frac{1}{2}(Tx_n - x_n) \to 0.$$

In addition, $\widehat{y} = \operatorname{prox}_{\gamma g}(\widehat{x}) \in \operatorname{zer}(\partial f + \partial g)$.
Since

$$\partial f(\widehat{y}) + \partial g(\widehat{y}) \subset \partial(f+g)(\widehat{y}),$$

$\widehat{y} \in \operatorname{zer}\big(\partial(f+g)\big) = \operatorname{Argmin}(f+g)$.

Douglas-Rachford algorithm

Let $\mathcal{H}$ be a Hilbert space.

Let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$.

Let $\gamma \in]0, +\infty[$ and let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[0, 2]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(2 - \lambda_n) = +\infty$.

We assume that $\operatorname{zer}(\partial f + \partial g) \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ z_n = \operatorname{prox}_{\gamma f}(2y_n - x_n) \\ x_{n+1} = x_n + \lambda_n(z_n - y_n). \end{cases}$$

The following properties are satisfied:

1. $x_n \rightharpoonup \widehat{x}$
2. $\widehat{y} = \operatorname{prox}_{\gamma g}\widehat{x} \in \operatorname{Argmin}(f + g)$
3. $z_n - y_n \to 0$
4. $y_n \rightharpoonup \widehat{y}$, $z_n \rightharpoonup \widehat{y}$.

Douglas-Rachford algorithm

Proof: For simplicity, assume that $\mathcal{H}$ is finite dimensional. $\mathrm{prox}_{\gamma g}$ being continuous (since nonexpansive), we have

$$y_n \to \mathrm{prox}_{\gamma g}\widehat{x} = \widehat{y}.$$

Because $z_n - y_n \to 0$, we deduce that $z_n \to \mathrm{prox}_{\gamma g}\widehat{x} = \widehat{y}$.

Douglas-Rachford algorithm

Let $\mathcal{H}$ be a Hilbert space.
Let $f \in \Gamma_0(\mathcal{H})$ and $g \in \Gamma_0(\mathcal{H})$.
Let $\gamma \in]0, +\infty[$ and let $(\lambda_n)_{n\in\mathbb{N}}$ be a sequence in $[0, 2]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(2 - \lambda_n) = +\infty$.
We assume that $\operatorname{zer}(\partial f + \partial g) \neq \varnothing$. Let $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ z_n = \operatorname{prox}_{\gamma f}(2y_n - x_n) \\ x_{n+1} = x_n + \lambda_n(z_n - y_n). \end{cases}$$

The following properties are satisfied:

1. $x_n \rightharpoonup \widehat{x}$
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Remark: The limit case when $(\forall n \in \mathbb{N})\ \lambda_n = 2$ is the Peaceman-Rachford algorithm. Its convergence is only guaranteed under some additional assumption (e.g. strong convexity of g).

Parallel form of Douglas-Rachford

Lemma

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces.
Let $L \in \mathcal{B}(\mathcal{G}, \mathcal{H})$.
Then L is weakly continuous.

Proof: Let $(x_n)_{n\in\mathbb{N}}$ be a sequence of $\mathcal{G}$ such that $x_n \rightharpoonup x \in \mathcal{G}$.
Then, for every $y \in \mathcal{H}$,

$$\begin{aligned} & \langle x_n \mid L^* y \rangle \to \langle x \mid L^* y \rangle \\ \Leftrightarrow \quad & \langle L x_n \mid y \rangle \to \langle L x \mid y \rangle . \end{aligned}$$

Thus $L x_n \rightharpoonup L x$.

Parallel form of Douglas-Rachford

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces.
Let $g \in \Gamma_0(\mathcal{H})$ and $L \in \mathcal{B}(\mathcal{G}, \mathcal{H})$ be such that L^*L is an isomorphism.
Let $\gamma \in \left]0, +\infty\right[$ and let $(\lambda_n)_{n\in\mathbb{N}}$ a sequence in $[0, 2]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(2 - \lambda_n) = +\infty$.
Assume that $\operatorname{zer}(L^* \circ \partial g \circ L) \neq \varnothing$. Let $x_0 \in \mathcal{H}$, $v_0 = (L^*L)^{-1}L^*x_0$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ c_n = (L^*L)^{-1}L^*y_n \\ x_{n+1} = x_n + \lambda_n\big(L(2c_n - v_n) - y_n\big) \\ v_{n+1} = v_n + \lambda_n(c_n - v_n). \end{cases}$$

Then $c_n \rightharpoonup \widehat{v}$ and $v_n \rightharpoonup \widehat{v}$ where $\widehat{v} \in \operatorname{Argmin}(g \circ L)$.

Parallel form of Douglas-Rachford

Proof: Let $E = \operatorname{ran} L$. E is closed.
Indeed, let $(Lw_n)_{n\in\mathbb{N}}$ be sequence of E converging to $z \in \mathcal{H}$.
By continuity,

$$\begin{aligned} L^*Lw_n \to L^*z \quad &\Rightarrow \quad w_n \to (L^*L)^{-1}L^*z = w \\ &\Rightarrow \quad Lw_n \to Lw. \end{aligned}$$

This shows that the limit of any convergent sequence of elements of E belongs to E.

Parallel form of Douglas-Rachford

Proof: Let $E = \operatorname{ran} L$. Since E is closed, $\iota_E \in \Gamma_0(\mathcal{H})$ and

$$
\begin{aligned}
\operatorname{zer}(L^* \circ \partial g \circ L) \neq \varnothing \quad &\Leftrightarrow \quad (\exists v \in \mathcal{G})\ 0 \in L^* \partial g(Lv) \\
&\Leftrightarrow \quad (\exists x \in E)\ 0 \in L^* \partial g(x) \\
&\Leftrightarrow \quad (\exists x \in E)(\exists u \in \partial g(x))\ \ 0 = L^* u \\
&\Leftrightarrow \quad (\exists x \in E)(\exists u \in \partial g(x))\ \ -u \in \operatorname{Ker} L^* = E^{\perp} \\
&\Leftrightarrow \quad (\exists x \in \mathcal{H})(\exists u \in \partial g(x))\ \ -u \in N_E(x) = \partial \iota_E(x) \\
&\Leftrightarrow \quad (\exists x \in \mathcal{H})\ 0 \in \partial \iota_E(x) + \partial g(x) \\
&\Leftrightarrow \quad \operatorname{zer}(\partial \iota_E + \partial g) \neq \varnothing.
\end{aligned}
$$

We next apply Douglas-Rachford algorithm with $f = \iota_E$.

Parallel form of Douglas-Rachford

Proof: $f = \iota_E \Rightarrow \operatorname{prox}_{\gamma f} = P_E$. The DR algorithm reads

$$(\forall n \in \mathbb{N}) \qquad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ z_n = P_E(2y_n - x_n) = 2P_E y_n - P_E x_n \\ x_{n+1} = x_n + \lambda_n(z_n - y_n). \end{cases}$$

For every $n \in \mathbb{N}$, $P_E y_n = L c_n$
with

$$\begin{aligned} c_n &= \operatorname*{argmin}_{c \in \mathcal{G}} \ \|y_n - Lc\|^2 \\ \Leftrightarrow \quad & L^*(Lc_n - y_n) = 0 \\ \Leftrightarrow \quad & c_n = (L^*L)^{-1} L^* y_n. \end{aligned}$$

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For every $n \in \mathbb{N}$, $P_E y_n = L c_n$
with $c_n = (L^*L)^{-1}L^* y_n$
and $P_E x_n = L v_n$ with $v_n = (L^*L)^{-1}L^* x_n$.
In addition

$$\begin{aligned} L v_{n+1} = P_E x_{n+1} &= P_E x_n + \lambda_n(P_E z_n - P_E y_n) \\ &= P_E x_n + \lambda_n(z_n - P_E y_n) \\ &= P_E x_n + \lambda_n(P_E y_n - P_E x_n) \\ &= L(v_n + \lambda_n(c_n - v_n)), \end{aligned}$$

which yields $v_{n+1} = v_n + \lambda_n(c_n - v_n)$.

Parallel form of Douglas-Rachford

Proof: In summary

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ c_n = (L^*L)^{-1} L^* y_n \\ z_n = L(2c_n - v_n) \\ x_{n+1} = x_n + \lambda_n (z_n - y_n) \\ v_{n+1} = v_n + \lambda_n (c_n - v_n). \end{cases}$$

We know that

$$\begin{aligned} y_n \rightharpoonup \widehat{y} &\in \operatorname{Argmin}(\iota_E + g) \\ &\Leftrightarrow \widehat{y} = L\widehat{v} \text{ and } (\forall y \in E)\ g(y) \geq g(\widehat{y}) \\ &\Leftrightarrow \widehat{y} = L\widehat{v} \text{ and } (\forall v \in \mathcal{G})\ g(Lv) \geq g(L\widehat{v}) \\ &\Leftrightarrow \widehat{y} = L\widehat{v} \text{ and } \widehat{v} \in \operatorname{Argmin}(g \circ L). \end{aligned}$$

Parallel form of Douglas-Rachford

Proof: In summary

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ c_n = (L^*L)^{-1} L^* y_n \\ z_n = L(2c_n - v_n) \\ x_{n+1} = x_n + \lambda_n (z_n - y_n) \\ v_{n+1} = v_n + \lambda_n (c_n - v_n). \end{cases}$$

We know that

$$y_n \rightharpoonup \widehat{y} = L\widehat{v} \text{ and } \widehat{v} \in \operatorname{Argmin}(g \circ L).$$

Because of the weak continuity of P_E,

$$P_E y_n = L c_n \rightharpoonup P_E \widehat{y} = \widehat{y} = L\widehat{v}$$

and, by using the weak continuity of $(L^*L)^{-1}L^*$,

$$c_n \rightharpoonup \widehat{v} \in \operatorname{Argmin}(g \circ L).$$

Parallel form of Douglas-Rachford

Proof: In summary

$$(\forall n \in \mathbb{N}) \qquad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ c_n = (L^*L)^{-1} L^* y_n \\ z_n = L(2c_n - v_n) \\ x_{n+1} = x_n + \lambda_n (z_n - y_n) \\ v_{n+1} = v_n + \lambda_n (c_n - v_n). \end{cases}$$

In addition

$$z_n = L(2c_n - v_n) \rightharpoonup \widehat{y} = L\widehat{v}$$

and, because of the weak continuity of $(L^*L)^{-1}L^*$,

$$2c_n - v_n \rightharpoonup \widehat{v} \quad \Rightarrow \quad v_n \rightharpoonup \widehat{v}.$$

Parallel form of Douglas-Rachford

Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces.
Let $g \in \Gamma_0(\mathcal{H})$ and $L \in \mathcal{B}(\mathcal{G}, \mathcal{H})$ be such that $\operatorname{ran} L$ is closed and L^*L is an isomorphism.
Let $\gamma \in \left]0, +\infty\right[$ and let $(\lambda_n)_{n\in\mathbb{N}}$ a sequence in $[0, 2]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(2 - \lambda_n) = +\infty$.
Assume that $\operatorname{zer}(L^* \circ \partial g \circ L) \neq \varnothing$. Let $x_0 \in \mathcal{H}$, $v_0 = (L^*L)^{-1}L^*x_0$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_n = \operatorname{prox}_{\gamma g} x_n \\ c_n = (L^*L)^{-1}L^*y_n \\ x_{n+1} = x_n + \lambda_n\big(L(2c_n - v_n) - y_n\big) \\ v_{n+1} = v_n + \lambda_n(c_n - v_n). \end{cases}$$

Then $v_n \rightharpoonup \widehat{v}$ where $\widehat{v} \in \operatorname{Argmin}(g \circ L)$.

Question: Give the particular case of the algorithm when
$\mathcal{H} = \mathcal{H}_1 \times \cdots \times \mathcal{H}_m$
$\big(\forall x = (x_1, \ldots, x_m) \in \mathcal{H}\big)\ g(x) = \sum_{i=1}^m g_i(x_i)$
$L\colon v \mapsto (L_1 v, \ldots, L_m v)$.

Parallel form of Douglas-Rachford

PPXA+

Let $\mathcal{H}_1, \ldots, \mathcal{H}_m$ and $\mathcal{G}$ be Hilbert spaces.

For every $i \in \{1, \ldots, m\}$, let $g_i \in \Gamma_0(\mathcal{H}_i)$ and let $L_i \in \mathcal{B}(\mathcal{G}, \mathcal{H}_i)$.

Assume that $\sum_{i=1}^m L_i^* L_i$ is an isomorphism.

Let $\gamma \in \left]0, +\infty\right[$ and let $(\lambda_n)_{n\in\mathbb{N}}$ a sequence in $[0, 2]$ such that $\sum_{n\in\mathbb{N}} \lambda_n(2 - \lambda_n) = +\infty$. Assume that $\operatorname{zer}\left(\sum_{i=1}^m L_i^* \circ \partial g_i \circ L_i\right) \neq \varnothing$.

Let $(x_{0,i})_{1\leq i\leq m} \in \mathcal{H}_1 \times \cdots \times \mathcal{H}_m$, $v_0 = \left(\sum_{i=1}^m L_i^* L_i\right)^{-1} \sum_{i=1}^m L_i^* x_{0,i}$ and

$$(\forall n \in \mathbb{N}) \quad \begin{cases} y_{n,i} = \operatorname{prox}_{\gamma g_i}(x_{n,i}), \quad i \in \{1, \ldots, m\} \\ c_n = \left(\sum_{i=1}^m L_i^* L_i\right)^{-1} \sum_{i=1}^m L_i^* y_{n,i} \\ x_{n+1,i} = x_{n,i} + \lambda_n\big(L_i(2c_n - v_n) - y_{n,i}\big), \quad i \in \{1, \ldots, m\} \\ v_{n+1} = v_n + \lambda_n(c_n - v_n). \end{cases}$$

Then $v_n \rightharpoonup \widehat{v} \in \operatorname{Argmin} \sum_{i=1}^m g_i \circ L_i$.

Exercise

Let $\mathcal{G}$ be a real Hilbert space. Let $m \in \mathbb{N}$, $m \geq 2$.

1. For every $i \in \{1, \ldots, m\}$, let g_i be a function in $\Gamma_0(\mathcal{G})$ such that $\operatorname{dom} g_1 \cap \left(\bigcap_{i=2}^{m} \operatorname{int} (\operatorname{dom} g_i) \right) \neq \varnothing$.
   Propose a parallel algorithm for finding a minimizer of $\sum_{i=1}^{m} g_i$ when such a minimizer exists.
2. Let $p \in \{1, \ldots, m-1\}$ and let $(C_i)_{1 \leq i \leq m}$ be nonempty closed convex subsets of $\mathcal{G}$. We assume that $\bigcap_{i=1}^{p} \operatorname{int} (C_i) \neq \varnothing$, and for every $j \in \{p+1, \ldots, m\}$, C_j is bounded.
   Propose a parallel algorithm for solving the problem

$$\underset{x \in \bigcap_{i=1}^{p} C_i}{\operatorname{minimize}} \sum_{j=p+1}^{m} \sigma_{C_j}(x)$$

   when this problem admits a solution.
3. Let $\mathcal{H} = \mathbb{R}^N$. Deduce an algorithm to

$$\underset{x \in \bigcap_{i=1}^{p} C_i}{\operatorname{minimize}} \ \|x\|_1.$$